ASCII MASTER FLOW — PANEL 1/8: Endogenic system: plate motion, strain release and magma generation

```
+-------------------- ENDOGENIC SYSTEM --------------------+
| Internal heat + gravity + plate motion reshape crust.  |
+----------------------------------------------------------+
                         |
           +-------------+-------------+
           |                           |
     STRESS ACCUMULATION         MAGMA GENERATION
           |                           |
 load -> lock -> strain       divergence: decompression
           |                  subduction: volatile flux
 rupture -> slip              plume: anomalous upwelling
           |                           |
      EARTHQUAKE                   VULCANISM
           \                           /
            +---- landforms + hazards + resources ----+

FAULT PAIRS
tension -> normal | compression -> reverse/thrust | shear -> strike-slip

Plate motion loads a region; sudden fault rupture is the immediate quake source.
```

ASCII MASTER FLOW — PANEL 2/8: Earthquake source geometry, depth and seismogram logic

```
                                SURFACE
station A -------- epicentre -------- station B
                        |
                        |
                 focus / hypocentre
              rupture begins on fault plane

SEISMOGRAM ARRIVAL ORDER
P first -> S second -> Love and Rayleigh later
     |
P-S interval at one station gives source distance
     |
three or more distance circles estimate the epicentre

DEPTH CLASS
shallow: 0-70 km | intermediate: 70-300 km | deep: 300-700 km

SEQUENCE FIREWALL
foreshock, mainshock and aftershock are relational labels;
a swarm has no single clearly dominant event.

Epicentre is a vertical projection, not necessarily the worst-damaged place.
```

ASCII MASTER FLOW — PANEL 3/8: Seismic waves, shadow zones, magnitude and intensity

WAVE	MOTION	MEDIUM	MAIN CLUE
P	compression	solid, liquid	core refraction
S	transverse shear	solid only	liquid outer core
Love	horizontal shear	surface	lateral shaking
Rayleigh	rolling ellipse	surface	mixed motion

P shadow about 103-142 deg | direct S shadow beyond about 103 deg

MAGNITUDE: one source measure; Mw uses seismic moment
moment = rigidity x fault area x average slip
+1 local magnitude gives about 10x amplitude and 31.6x energy

INTENSITY: place-specific observed effects; many values for one event

source x path x site x exposure x vulnerability / capacity -> damage

ASCII MASTER FLOW — PANEL 4/8: Global seismic belts, cascading hazards and tsunami sequence

```
GLOBAL EARTHQUAKE DISTRIBUTION
+----------------------+----------------------+--------------------+
| Circum-Pacific       | Mediterranean-       | Ridges/transforms  |
| trench + slab + arc  | Himalayan collision | shallow events     |
| megathrust + tsunami | complex convergence | basaltic ridges    |
+----------------------+----------------------+--------------------+
Intraplate events reactivate inherited structures; stable is not aseismic.

CASCADE
shaking -> amplification / liquefaction / lateral spreading
        -> landslide / fire / lifeline failure / dam distress

TSUNAMI
submarine megathrust -> vertical sea-floor displacement
-> long low deep-ocean wave -> coastal shoaling
-> run-up + inundation + return flow

First wave need not be largest; withdrawal need not occur before every event.
Risk varies with bathymetry, bay shape, shelf, exposure and evacuation.
```

ASCII MASTER FLOW — PANEL 5/8: Volcano anatomy, magma controls, landforms and eruption styles

```
                       ash plume
                           |
                          /\
                         /  \ crater
                  lava --    -- lava
                         |vent|
                dyke ----|    |---- sill
                         |conduit
                    magma chamber
```

Magma below surface | lava after eruption
Silica up -> viscosity up -> degassing harder -> explosivity may rise
Temperature up -> viscosity generally down
Retained gas + decompression -> bubbles -> fragmentation

FORMS: batholith, laccolith, lopolith, phacolith, dyke, sill
Dyke cuts layers; sill follows layers.

STYLE: fissure/Icelandic -> Hawaiian -> Strombolian -> Vulcanian -> Plinian
FORM: shield | composite | cinder cone | dome | fissure plateau
STATUS: active | dormant | extinct; status is not a compulsory life cycle.

ASCII MASTER FLOW — PANEL 6/8: Volcanic settings, global belts, products and environmental effects

SETTING	MELTING MECHANISM	EXPRESSION
Divergent	decompression	ridge, rift, basalt
Subduction	slab volatile flux	trench-slab-arc system
Plume	mantle upwelling	flood basalt, hot spot

Ring of Fire = trench + descending slab + earthquakes + volcanic arc
Plume model: broad head may form flood basalt; tail may form age chain.
Deccan Trap is a fissure-fed flood-basalt plateau, not a giant cone.

HAZARDS: lava | ash | density currents | lahars | gases
CLIMATE: stratospheric sulphate aerosol can cause temporary cooling
BENEFITS: new crust/land | soil parent | geothermal energy | minerals

Impact converges through wind, relief, drainage, exposure and preparedness.

ASCII MASTER FLOW — PANEL 7/8: India seismic geography, zonation, institutions and resilience stack

```
INDIA REGIONAL HIERARCHY
Himalaya        : collision, thrusts, slopes, dense corridors
North-East      : convergence, strike-slip complexity
Andaman         : subduction, tsunami, Barren Island volcanism
Kachchh         : inherited rift and fault reactivation
Peninsula       : relatively stable; Koyna and Latur show residual risk
Indo-Gangetic   : alluvium, amplification and high exposure
                         |
MACROZONATION: broad design hazard, not prediction
MICROZONATION: geology + soil + motion + water + liquefaction + GIS
                         |
NCS/MoES -> monitoring | GSI -> geology | BIS -> standards
NDMA/NIDM -> guidance | INCOIS -> tsunami warning
States/ULBs -> approvals + inspection + enforcement
                         |
site -> regular load path -> ductility -> connections
-> non-structural safety -> quality control -> retrofit -> drills

Early warning detects rupture after it starts; it is not prediction.
```

ASCII MASTER FLOW — PANEL 8/8: Hazard-risk-response and Mains answer spine for seismic India

```
QUESTION: Why can similar earthquakes produce unequal disasters?
                             |
HAZARD: source size + depth + rupture geometry + path
                             |
SITE: basin sediment + groundwater + slope + liquefaction potential
                             |
EXPOSURE: people + buildings + lifelines + time of occurrence
                             |
VULNERABILITY: weak design + irregular form + poor maintenance
                             |
CAPACITY: codes + enforcement + retrofit + warning + response
                             |
RISK VERDICT: hazard becomes disaster through social and spatial filters

15-MARK INDIA SPINE
tectonic regions -> site modifiers -> exposure/vulnerability -> cases
-> zonation and microzonation -> construction and governance -> conclusion

TRAPS: focus is not epicentre | magnitude is not intensity
tsunami is not a tide | caldera is not an ordinary crater
zone map is not prediction | code publication is not enforcement.
```